On-Policy Prediction with Approximation

Unit - 2

Reinforcement Learning &
Game Theory

Summary

- Introduction
- Value-function approximation
- The Prediction Objective
- Stochastic-Gradient and Semi-Gradient Methods
- Linear Value Function Approximation
- Feature Construction for Linear Methods (Hints)
- Nonlinear Value Function Approximation (Hints)

Introduction

Revise from previous lectures

- So far we have represented value function by a lookup table
  - Every state s has an entry V(s), or
  - Every state-action pair (s,a) has an entry Q(s,a)
- Problem with large MDPs:
  - There are too many states and/or actions to store in memory
  - It is too slow to learn the value of each state individually
  -

Introduction

- **Problem**: **How can we extend tabular RL methods to apply them to problems with arbitrarily large state spaces?**
- E.g., # possible camera images > # atoms in the universe

- Almost all **states** encountered have **never been seen before** → **Generalization from previously encountered (similar) states**
  - Solution:
    - Estimate value function with function approximation

Parameterized form →

$$\hat{v}(s, \mathbf{w}) \approx v_\pi(s)$$

$$\text{or } \hat{q}(s, a, \mathbf{w}) \approx q_\pi(s, a)$$

Generalize from seen states to unseen states

- **Combine RL with function approximation (supervised learning)**

- The **RL setting** introduces **new issues** to **supervised learning**: e.g., nonstationarity, bootstrapping, delayed targets

Introduction

- **Goal / Solution**: substitute **tabular representations** of **state-value function** v_π with **function approximations**

- Approximated v_π are **estimated** from **on-policy** data, i.e., from experience generated using the known policy π.

- Approximations are based on **parametrized functions** $\hat{v}(s,\mathbf{w}) \approx v_\pi(s)$ where $\mathbf{w}$ is the vector of parameters (known as weight vector)

- Example: $\hat{v}$ might be a **linear function** in features of the state with $\mathbf{w}$ vector of feature weights, or a multilayer **artificial neural network** with $\mathbf{w}$ vector of connection weights, or a **decision tree** with $\mathbf{w}$ split points and leaf values of the tree

- **Adjusting the weights** several functions can be implemented

Introduction

- Typically, the number of weights is much less than the number of states, i.e., $d \ll |\mathcal{S}|$

- Changing **one weight** changes the estimated value of **many states** (**generalization**)

- **Generalization** makes **reinforcement learning** more **powerful** but also more **difficult** to manage and understand

- Extending RL to function approximation makes it applicable to **partially observable problems** (i.e., full state not available)

Value-function approximation

Value-function approximation

- All **prediction methods** seen so far are based on **updates** of an
  estimated value function
  s → u means that the estimated value for state s should be more like the update target u
  MC update: $S_t \mapsto G_t$

  State updated

  Update target

- TD update: $S_t \mapsto \underbrace{R_{t+1} + \gamma \hat{v}(S_{t+1}, \mathbf{w}_t)}$

  State updated

  Update target

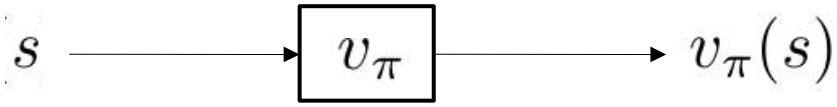
- DP update: $s \mapsto \underbrace{\mathbb{E}_\pi[R_{t+1} + \gamma \hat{v}(S_{t+1}, \mathbf{w}_t) \mid S_t = s]}$

  State updated

  Update target

Value-function approximation

- Can we interpret each **update** as specifying an **example** of the **desired input-output behaviour of the value function? Yes!**



- **Update in tabular representations** of the value function: the table entry for the **estimated value of state *s*** is **shifted** a fraction of the way **towards** the **target *u*** (**estimated values of other states are unchanged**)

- **Update in function approximations** of the value function: **arbitrary complex parameter updates** are available. **Updating at state *s* can change value estimations of other states**

- **Supervised learning** can be used to compute weights $\mathbf{w}$ using **function approximation** methods

Value-function approximation

- **Problem:** **not all** function approximation methods are **equally well suited for RL**

- In RL, learning must be performed **online**, while the agent interacts with the environment.

- We need **learning methods** that
  - learn efficiently from **incrementally acquired data**
  - handle **nonstationary target functions**

- Example: in GPI we seek to learn q_π as π changes.

- Methods that cannot deal with such nonstationarity are less suitable for RL

The Prediction Objective

The Prediction Objective

- Which **objective** do we use to **evaluate** the **approximated function**?

- In **tabular case** a continuous measure of prediction quality was not necessary because
  - the learned value function could become equal to the true one
  - updates affect only single states

- With **function approximation** these two assumptions are not guaranteed
- We define **which states we care most about** defining a **state distribution** $\mu(s) \geq 0, \sum_s \mu(s) = 1$

- By the error in a state '*s*' we mean the square of the difference between the approximate value $\hat{v}(s,w)$ and the true value $v_\pi(s)$. Weighting this over the state space by μ, we obtain a natural objective function, the Mean Squared Value Error, denoted VE:

$$\boxed{\overline{\mathrm{VE}}(\mathbf{w}) \doteq \sum_{s \in \mathcal{S}} \mu(s) \Big[v_\pi(s) - \hat{v}(s,\mathbf{w}) \Big]^2}$$

The Prediction Objective

- The square root of $\overline{\mathrm{VE}}$ provides a **measure** of how much the **approximate** values differ form the **true** values

- Often $\mu(s)$ is set to the fraction of time spent in state *s* (***on-policy distribution***)

On Policy distribution in episodic tasks

- In episodic tasks: let *h(s)* the probability an episode starts in state *s*, then the **number of time steps spent, on average, in state s in a single episode** is

$$\eta(s) = h(s) + \sum_{\bar{s}} \eta(\bar{s}) \sum_{a} \pi(a|\bar{s}) p(s|\bar{s}, a), \quad \text{for all } s \in \mathcal{S}$$

and the on-policy distribution is then

$$\mu(s) = \frac{\eta(s)}{\sum_{s'} \eta(s')}, \quad \text{for all } s \in \mathcal{S}$$

The Prediction Objective

- In **continuing tasks** the on-policy distribution is the stationary distribution under π. In **episodic tasks** also depends on state initial probability

- The **formal analysis** of the **continuing** and **episodic** cases must be treated separately with value function approximation

The Prediction Objective

- The reason we are learning a value function–is to find a better policy. The best value function for this purpose is not necessarily the best for minimizing VE.
- The **goal** of $\overline{\mathrm{VE}}$ is to find a **global optimum**, namely, a **weight vector** $\mathbf{w}^*$ for which

$$\overline{\mathrm{VE}}(\mathbf{w}^*) \leq \overline{\mathrm{VE}}(\mathbf{w})$$

  for all possible $\mathbf{w}$

- This is **possible** for **simple function approximators** (e.g., linear models) rarely for **complex approximators** (e.g., ANNs and decision trees) in which learning usually converges to **local optima**, i.e., $\mathbf{w}^*$ for which

$$\overline{\mathrm{VE}}(\mathbf{w}^*) \leq \overline{\mathrm{VE}}(\mathbf{w})$$

  for all $\mathbf{w}$ **in some neighbourhood of** $\mathbf{w}^*$

- This is the **best** that can be done and it is usually **enough** although in many cases there is **no guarantees of convergence to the optimum**

The Prediction Objective

In summary, so far we have described:

- A **framework** for **combining** RL methods for **value prediction** with **function approximation** methods (using RL updates as training examples)

- A $\overline{\mathrm{VE}}$ performance measure that these methods may aspire to minimize

In the rest of the lecture we will consider **function approximation methods** based on **gradient-descent** since they are particularly **promising** and reveal **key theoretical properties**

Stochastic-Gradient and Semi-Gradient Methods

Stochastic-gradient Methods (SDG)

- **Class of learning methods** for function approximation in value prediction: **Stochastic Gradient Descent (SGD)**
- Among the most **widely used** of all function approximation methods
- Well suited to **online RL**

Let:

- $\mathbf{w} \doteq (w_1, w_2, \ldots, w_d)^\top$, where w is a column vector with fixed number of real valued components
- $\hat{v}(s,\mathbf{w})$ is a **differentiable** function of $\mathbf{w}$ for all states *s*

At each time step $t = 0, 1, 2, 3, \ldots,$ we observe a new example $S_t \mapsto v_\pi(S_t)$and update $\mathbf{w}_t$

States S_t can be **randomly selected** or they can be **successive** states of an interaction with the environment

Stochastic-gradient Methods (SGD)

- Values $v_\pi(S_t)$ are **unknown** but even though we could observe their **true values**, learning the approximate function would be difficult

- The approximator has **limited "resolution"**. There is no $\mathbf{w}$ that gets all the states exactly correct

- **Goal of SGD: to minimize error i.e. VE on the observed examples**

- **Strategy of SGD: adjust $\mathbf{w}$ after each example by a small amount in the direction that would most reduce the error on that example**

$$\begin{aligned}\mathbf{w}_{t+1} &\doteq \mathbf{w}_t - \frac{1}{2}\alpha\nabla\Big[v_\pi(S_t) - \hat{v}(S_t,\mathbf{w}_t)\Big]^2 \\ &= \mathbf{w}_t + \alpha\Big[v_\pi(S_t) - \hat{v}(S_t,\mathbf{w}_t)\Big]\nabla\hat{v}(S_t,\mathbf{w}_t),\end{aligned}$$

where α >0 and $\nabla f(\mathbf{w}) \doteq \left(\frac{\partial f(\mathbf{w})}{\partial w_1}, \frac{\partial f(\mathbf{w})}{\partial w_2}, \ldots, \frac{\partial f(\mathbf{w})}{\partial w_d}\right)^\top$ **gradient** of *f w.r.t* ***w***

Stochastic-gradient Methods (SGD)

- The **negative gradient of the example's squared error** is the **direction in which the error falls most rapidly**

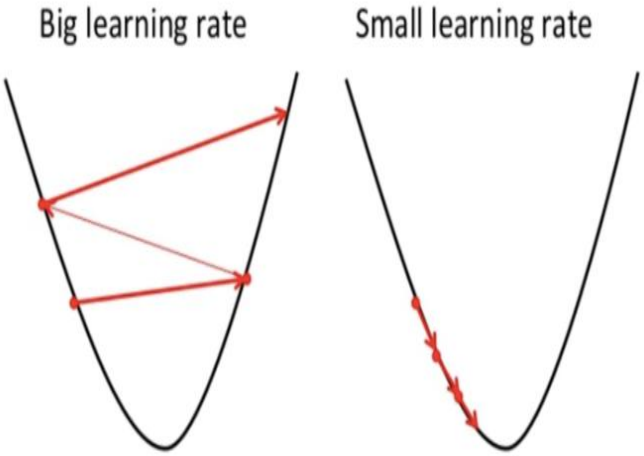
- SDG is called **"stochastic"** when the **update is done on only a single sample.**

- **Here, the term "stochastic" comes from the fact that the gradient based on a single training sample is a "stochastic approximation" of the "true" cost gradient.**

- Over many examples, making small steps, α the effect is to minimize $\overline{\mathrm{VE}}$

- Why performing only **"small" steps**?

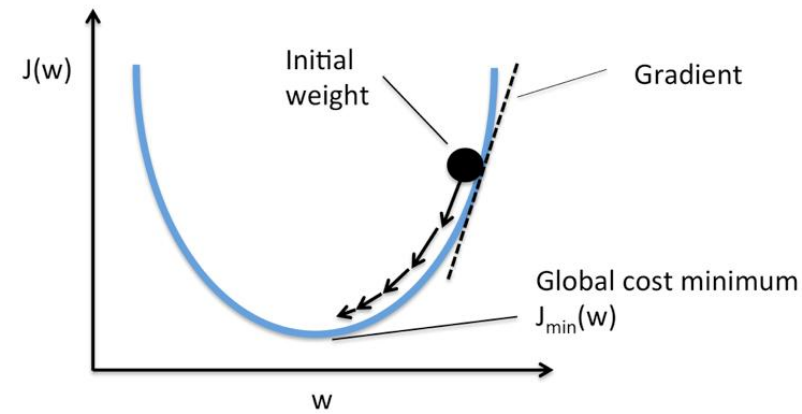
If we **completely corrected each example in one step** then we would **not balance the error** (which cannot be completely removed) **on all samples**



Flashback of gradient Methods (GD)

- Using the Gradient Decent optimization algorithm, the weights are updated incrementally after each epoch (= pass over the training dataset).
- The magnitude and direction of the weight update is computed by taking a step in the opposite direction of the cost gradient

$$\Delta w_j = -\eta \frac{\partial J}{\partial w_j},$$



- where η is the learning rate. The weights are then updated after each epoch via the following update rule:

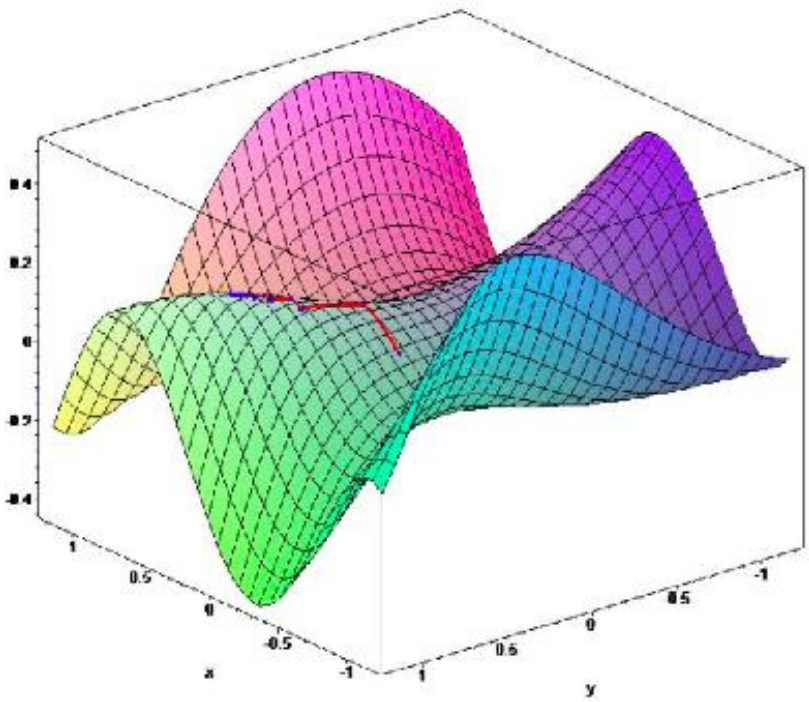
$$w := w + \Delta w,$$

- In Gradient Descent optimization, we compute the cost gradient based on the complete training set while in SGD, the gradient based on a single training sample.

Flashback of gradient Methods (GD)-1

- Let J(w) be a differentiable function of parameter vector w
- Define the gradient of J(w) to be:

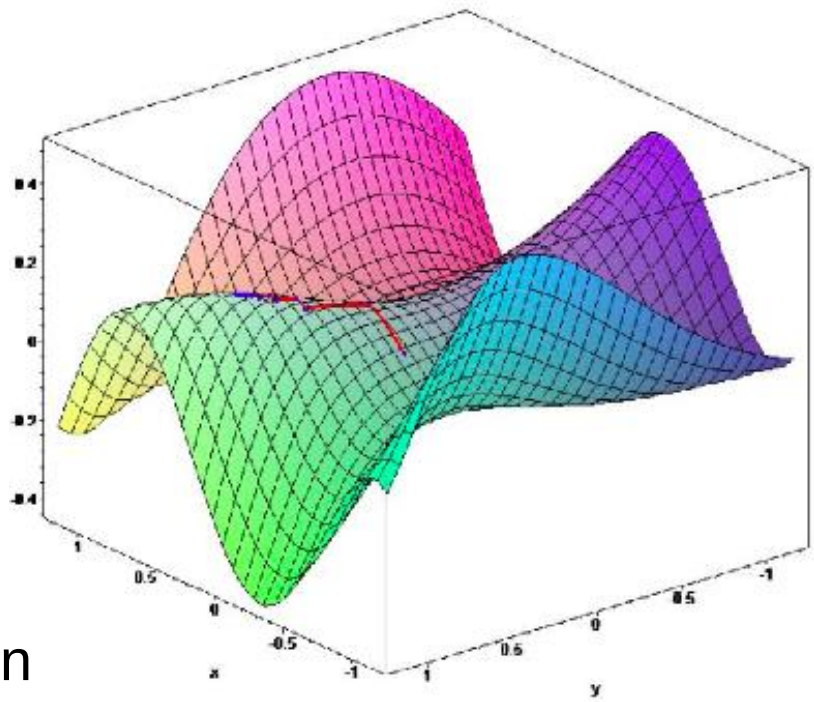
$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \begin{pmatrix} \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}_1} \\ \vdots \\ \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}_n} \end{pmatrix}$$



Flashback of gradient Methods (GD)-2

- Let J(w) be a differentiable function of parameter vector w
- Define the gradient of J(w) to be:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \begin{pmatrix} \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}_1} \\ \vdots \\ \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}_n} \end{pmatrix}$$



- To find a local minimum of J(w), adjust w in direction of the negative gradient:

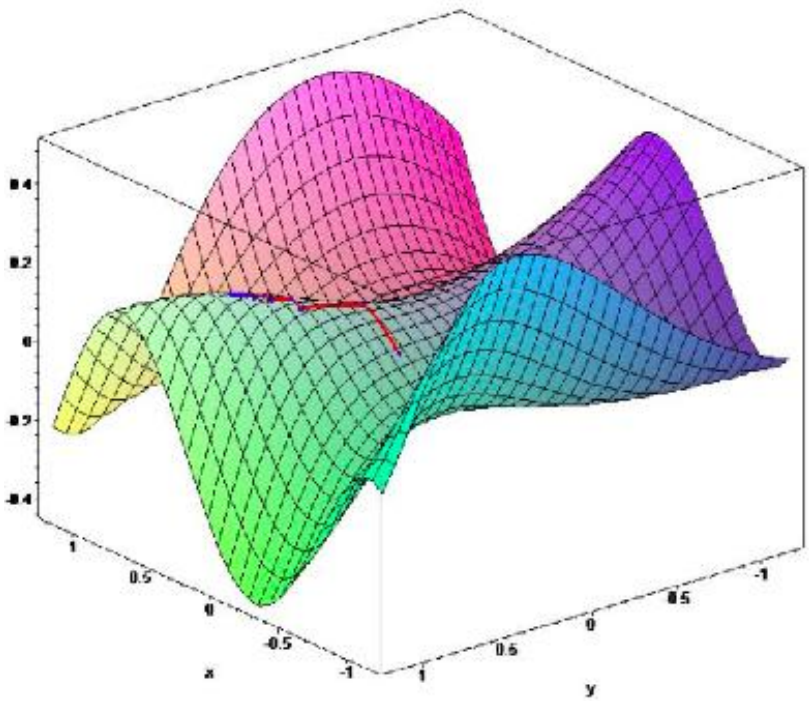
$$\Delta \mathbf{w} = -\frac{1}{2}\alpha \nabla_{\mathbf{w}} J(\mathbf{w})$$

Step-size

Flashback of gradient Methods (GD)-3

- Let J(w) be a differentiable function of parameter vector w
- Define the gradient of J(w) to be:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \begin{pmatrix} \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}_1} \\ \vdots \\ \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}_n} \end{pmatrix}$$



- Starting from a guess $\mathbf{w}_0$
- We consider the sequence $\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2, \ldots$ s.t. :

$$\mathbf{w}_{n+1} = \mathbf{w}_n - \frac{1}{2}\alpha \nabla_{\mathbf{w}} J(\mathbf{w}_n)$$

- We then have $J(\mathbf{w}_0) \geq J(\mathbf{w}_1) \geq J(\mathbf{w}_2) \geq \ldots$

Our objective

- Goal: find parameter vector w minimizing mean-squared error between the true value function $v_\pi(S)$ and its approximation $\hat{v}(S, \mathbf{w})$

$$J(\mathbf{w}) = \mathbb{E}_\pi \left[\left(v_\pi(S) - \hat{v}(S, \mathbf{w}) \right)^2 \right]$$

Gradient Descent

- Goal: find parameter vector w minimizing mean-squared error between the true value function $v_\pi(S)$ and its approximation $\hat{v}(S, \mathbf{w})$

$$J(\mathbf{w}) = \mathbb{E}_\pi \left[(v_\pi(S) - \hat{v}(S, \mathbf{w}))^2 \right]$$

$$\nabla_w J(w) = \mathbb{E}_\pi \left[2(v_\pi(S) - \hat{v}(S, w))(-\nabla_w \hat{v}(S, w)) \right]$$

Simplifying:

$$\nabla_w J(w) = -2\mathbb{E}_\pi \left[(v_\pi(S) - \hat{v}(S, w)) \nabla_w \hat{v}(S, w) \right]$$

$$\begin{aligned} \Delta \mathbf{w} &= -\frac{1}{2}\alpha \nabla_{\mathbf{w}} J(\mathbf{w}) \\ &= \alpha \mathbb{E}_\pi \left[(v_\pi(S) - \hat{v}(S, \mathbf{w})) \nabla_{\mathbf{w}} \hat{v}(S, \mathbf{w}) \right] \end{aligned}$$

Gradient Descent

- Goal: find parameter vector w minimizing mean-squared error between the true value function $v_\pi(S)$ and its approximation $\hat{v}(S, \mathbf{w})$

$$J(\mathbf{w}) = \mathbb{E}_\pi \left[(v_\pi(S) - \hat{v}(S, \mathbf{w}))^2 \right]$$

$$\begin{aligned} \Delta \mathbf{w} &= -\frac{1}{2} \alpha \nabla_{\mathbf{w}} J(\mathbf{w}) \\ &= \alpha \mathbb{E}_\pi \left[(v_\pi(S) - \hat{v}(S, \mathbf{w})) \nabla_{\mathbf{w}} \hat{v}(S, \mathbf{w}) \right] \end{aligned}$$

- Starting from a guess w_0

Gradient Descent

- Goal: find parameter vector w minimizing mean-squared error between the true value function $v_\pi(S)$ and its approximation $\hat{v}(S, \mathbf{w})$

$$J(\mathbf{w}) = \mathbb{E}_\pi \left[(v_\pi(S) - \hat{v}(S, \mathbf{w}))^2 \right]$$

- Gradient descent finds a local minimum:

$$\begin{aligned} \Delta \mathbf{w} &= -\frac{1}{2} \alpha \nabla_{\mathbf{w}} J(\mathbf{w}) \\ &= \alpha \mathbb{E}_\pi \left[(v_\pi(S) - \hat{v}(S, \mathbf{w})) \nabla_{\mathbf{w}} \hat{v}(S, \mathbf{w}) \right] \end{aligned}$$

Stochastic Gradient Descent

- Goal: find parameter vector w minimizing mean-squared error between the true value function $v_\pi(S)$ and its approximation $\hat{v}(S, \mathbf{w})$

$$J(\mathbf{w}) = \mathbb{E}_\pi \left[(v_\pi(S) - \hat{v}(S, \mathbf{w}))^2 \right]$$

- Gradient descent finds a local minimum:

$$\begin{aligned} \Delta \mathbf{w} &= -\frac{1}{2} \alpha \nabla_{\mathbf{w}} J(\mathbf{w}) \\ &= \alpha \mathbb{E}_\pi \left[(v_\pi(S) - \hat{v}(S, \mathbf{w})) \nabla_{\mathbf{w}} \hat{v}(S, \mathbf{w}) \right] \end{aligned}$$

- Stochastic gradient descent (SGD) samples the gradient:

$$\Delta \mathbf{w} = \alpha (v_\pi(S) - \hat{v}(S, \mathbf{w})) \nabla_{\mathbf{w}} \hat{v}(S, \mathbf{w})$$

Stochastic-gradient Methods (SGD)

- In practice the **target output observed** at time *t*, $U_t \in \mathbb{R}$, the tth training example, $S_t \rightarrow U_t$, is **not the true value** $v_\pi(S_t)$, but some **random approximation** of it (e.g., noisy corrupted value of $v_\pi(S_t)$ or a bootstrapping target)
- We perform an **approximate update** using $U_t \in \mathbb{R}$:

$$\mathbf{w}_{t+1} \doteq \mathbf{w}_t + \alpha\Big[U_t - \hat{v}(S_t,\mathbf{w}_t)\Big]\nabla\hat{v}(S_t,\mathbf{w}_t)$$

- If U_t is an **unbiassed estimate of the value**, i.e., $\mathbb{E}[U_t|S_t=s] = v_\pi(S_t)$ then $\mathbf{w}_t$ is guaranteed to converge to a local optimum

Gradient MC algorithm for estimating the value function

- The **Monte Carlo target** $U_t \doteq G_t$ is an unbiased estimate of $v_\pi(S_t)$, hence the **SGD version of MC state-value prediction converges**

Gradient Monte Carlo Algorithm for Estimating $\hat{v} \approx v_\pi$

Input: the policy π to be evaluated
Input: a differentiable function $\hat{v} : \mathcal{S} \times \mathbb{R}^d \to \mathbb{R}$
Algorithm parameter: step size $\alpha > 0$
Initialize value-function weights $\mathbf{w} \in \mathbb{R}^d$ arbitrarily (e.g., $\mathbf{w} = \mathbf{0}$)

Loop forever (for each episode):
  Generate an episode $S_0, A_0, R_1, S_1, A_1, \ldots, R_T, S_T$ using π
  Loop for each step of episode, $t = 0, 1, \ldots, T-1$:
    $\boxed{\mathbf{w} \leftarrow \mathbf{w} + \alpha\big[G_t - \hat{v}(S_t,\mathbf{w})\big]\nabla\hat{v}(S_t,\mathbf{w})}$

- Notice: MC provides a **non-bootstrapping estimate of** $v_\pi(S_t)$

Stochastic-gradient Methods (SGD)

- If a **bootstrapping estimate** of $v_\pi(S_t)$ is used as the target U_t (e.g., in TD and DP), then **convergence is not guaranteed**

- This is because the **target must be independent of** $\mathbf{w}_t$

- These methods are called **semi-gradient (bootstrapping) methods**

- They **do not converge as robustly as gradient methods** but they **converge reliably in important cases (e.g., linear case)**

- **Advantage of semi-gradient methods:**
  - They enable **faster learning**
  - They enable **learning continual and online**, **without waiting for the end of the episode**

Semi-Gradient TD algorithm for estimating the value function

Semi-gradient TD(0) for estimating $\hat{v} \approx v_\pi$

Input: the policy π to be evaluated
Input: a differentiable function $\hat{v} : \mathcal{S}^+ \times \mathbb{R}^d \to \mathbb{R}$ such that $\hat{v}(\text{terminal},\cdot) = 0$
Algorithm parameter: step size $\alpha > 0$
Initialize value-function weights $\mathbf{w} \in \mathbb{R}^d$ arbitrarily (e.g., $\mathbf{w} = \mathbf{0}$)

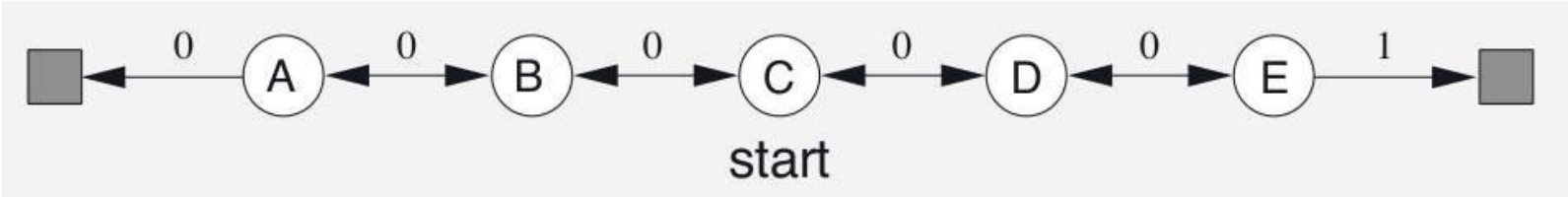
Loop for each episode:
  Initialize S
  Loop for each step of episode:
    Choose $A \sim \pi(\cdot|S)$
    Take action A, observe R, S'
    $\boxed{\mathbf{w} \leftarrow \mathbf{w} + \alpha\big[R + \gamma\hat{v}(S',\mathbf{w}) - \hat{v}(S,\mathbf{w})\big]\nabla\hat{v}(S,\mathbf{w})}$
    $S \leftarrow S'$
  until S is terminal

Example: state aggregation on the 1000-state random walk

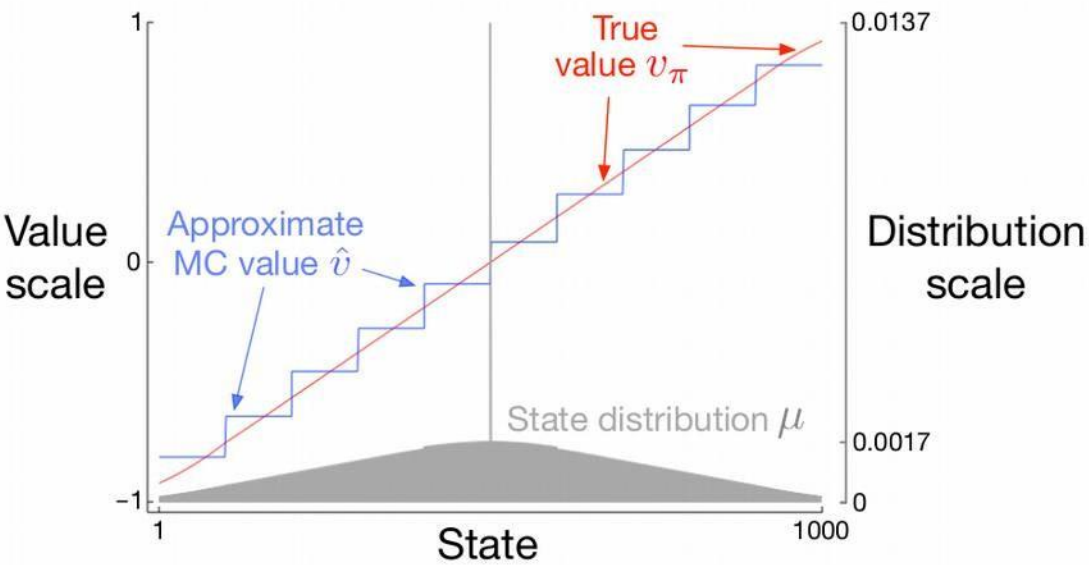
State aggregation: simple generalizing function approximation

- States are grouped together with one estimated value (constant)
- Each component of $\mathbf{w}$ is the estimation for a group of states
- The gradient $\nabla \hat{v}(S_t, \mathbf{w}_t)$ is 1 for the components of the group of S_t and 0 for the other components

- Consider a 1000-state version of of the random walk task



- **Function approximation by state aggregation using the gradient MC algorithm**
  - 100.000 episodes,
  - alpha=2x10^-5
  - 10 groups



Linear Value Function Approximation

Linear Methods

- Approximate function $\hat{v}(\cdot,\mathbf{w})$ with **linear function of the weight vector** $\mathbf{w}$

- For each state *s* there is a real-valued **feature vector**

$$\mathbf{x}(s) \doteq (x_1(s), x_2(s), \ldots, x_d(s))^\top$$

  with the same number of components (**features**) as $\mathbf{w}$ (i.e., *d*). The value of each feature is a **function of the state** $x_i : \mathcal{S} \to \mathbb{R}$

- **Linear method approximations of the state-value function** implement the **inner-product** between $\mathbf{w}$ and $\mathbf{x}(s)$:

$$\hat{v}(s,\mathbf{w}) \doteq \mathbf{w}^\top \mathbf{x}(s) \doteq \sum_{i=1}^{d} w_i x_i(s)$$

- The **approximate value-function** is said to be **linear in the weights**

Linear Methods

- It is natural to use **SGD updates** with **linear function approximations**

- The **gradient** of the approximate value function w.r.t. **w** in this case is

$$\nabla \hat{v}(s, \mathbf{w}) = \mathbf{x}(s)$$

- Hence, the **SDG update** becomes:

$$\mathbf{w}_{t+1} \doteq \mathbf{w}_t + \alpha \Big[U_t - \hat{v}(S_t, \mathbf{w}_t) \Big] \boxed{\mathbf{x}(S_t)}$$

- Simple form → Good for mathematical analysis (e.g., convergence)

- Only one optimum → local optimum = global optimum

Linear Methods

- The **Gradient Monte Carlo algorithm converges** to the global optimum of the $\overline{\mathrm{VE}}$ under linear function approximation

- The **semi-gradient TD(0) algorithm** also **converges** under linear function approximation. This result requires a separate theorem (the weight vector converges to a point near the local optimum)

  - The **update** at each time *t* is

$$\begin{aligned}\mathbf{w}_{t+1} &\doteq \mathbf{w}_t + \alpha\Big(R_{t+1} + \gamma \mathbf{w}_t^\top \mathbf{x}_{t+1} - \mathbf{w}_t^\top \mathbf{x}_t\Big)\mathbf{x}_t \\ &= \mathbf{w}_t + \alpha\Big(R_{t+1}\mathbf{x}_t - \mathbf{x}_t\big(\mathbf{x}_t - \gamma \mathbf{x}_{t+1}\big)^\top \mathbf{w}_t\Big)\end{aligned}$$

  - At steady state the expected next weight vector is

$$\mathbb{E}[\mathbf{w}_{t+1}|\mathbf{w}_t] = \mathbf{w}_t + \alpha(\mathbf{b} - \mathbf{A}\mathbf{w}_t)$$

with $\mathbf{b} \doteq \mathbb{E}[R_{t+1}\mathbf{x}_t] \in \mathbb{R}^d$ and $\mathbf{A} \doteq \mathbb{E}\Big[\mathbf{x}_t\big(\mathbf{x}_t - \gamma \mathbf{x}_{t+1}\big)^\top\Big] \in \mathbb{R}^d \times \mathbb{R}^d$

Linear Methods

- The **TD fixed point for linear semi-gradient TD(0)** can be computed as:

$$\begin{aligned} \mathbf{b} - \mathbf{A}\mathbf{w}_{\mathrm{TD}} &= \mathbf{0} \\ \Rightarrow \qquad \mathbf{b} &= \mathbf{A}\mathbf{w}_{\mathrm{TD}} \\ \Rightarrow \qquad \mathbf{w}_{\mathrm{TD}} &\doteq \mathbf{A}^{-1}\mathbf{b}. \end{aligned}$$

- At the TD fixed point it has been proved (in the continuing case) that the $\overline{\mathrm{VE}}$ **is within a bounded expansion of the lowest possible error**

$$\overline{\mathrm{VE}}(\mathbf{w}_{\mathrm{TD}}) \leq \frac{1}{1-\gamma} \min_{\mathbf{w}} \overline{\mathrm{VE}}(\mathbf{w})$$

- Namely, the **asimptotic error of the TD method** is no more than $\frac{1}{1-\gamma}$ times the smallest possible error, i.e., the error reached in the limit by the Monte Carlo method

- γ is usually close to 1 → **Substantial potential loss but TD methods have reduced variance and are faster than MC methods in practice**

References

- R. S. Sutton, A. G. Barto. Reinforcement learning, An Introduction. Second edition. Chapter 9